

Lecture 16

0/1 Knapsack Problem

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Problem

0/1 knapsack problem: Given items with defined **weights** and **values**, the objective is to **maximize total value** by selecting items for a knapsack without surpassing a **fixed capacity**. Each item must be either fully **included (1)** or **excluded (0)**.



Attribution: Generated by DALL-E, via ChatGPT (GPT-4), OpenAI, November 10, 2024.

Utilities

```
import requests

def read_knapsack_data(url):

    """
    Reads and processes knapsack problem data from a given URL.

    Args:
        url (str): The URL pointing to the data file.

    Returns:
        values, weights, capacity
```

Raises:

Exception: If there is an issue with fetching the data or parsing the content.

"""

try:

Fetch data from the URL
response = requests.get(url)

Raise an error if the request was unsuccessful
response.raise_for_status()

Split the data into lines
lines = response.text.strip().split('\n')

Parse the number of items
num_items = int(lines[0].strip())

Parse the values and weights lists
values = list(map(int, lines[1].strip().split()))
weights = list(map(int, lines[2].strip().split()))

Parse the capacity
capacity = int(lines[3].strip())

Validate that the number of items matches the length of values and weights
if len(values) != num_items or len(weights) != num_items:

raise ValueError("The number of items does not match the length of values or weights")

Return the values, weights, and capacity
return np.array(values), np.array(weights), capacity

except requests.exceptions.RequestException as e:

print(f"Error fetching data from URL: {e}")
raise

except ValueError as e:

print(f"Error processing data: {e}")
raise

except Exception as e:

print(f"An unexpected error occurred: {e}")
raise

Greedy Algorithms

” [Skiena:2008aa, page 192]

Greedy algorithms make the decision of what to do next by selecting the best local option from all available choices without regard to the global structure.

Greedy by weight

A possible greedy strategy involves selecting items in increasing order of weight until the total exceeds the capacity.

```
def greedy_knapsack_weight(weights, values, capacity):  
  
    """  
    Greedy algorithm for the 0/1 Knapsack Problem based on weight.  
  
    Args:  
        weights (np.ndarray): Weights of the items.  
        values (np.ndarray): Values of the items.  
        capacity (int): Capacity of the knapsack.  
  
    Returns:  
        tuple: Selected items (binary array), total value, total weight.  
    """  
  
    num_items = len(weights)  
  
    # Create a list of items with their values and original indices  
    items = list(zip(weights, values, range(num_items)))  
  
    # Sort items by weight in increasing order  
    items.sort()  
  
    total_weight = 0  
    total_value = 0  
    solution = np.zeros(num_items, dtype=int)
```

```

# Select items based on the sorted order
for w, v, idx in items:
    if total_weight + w <= capacity:
        solution[idx] = 1
        total_weight += w
        total_value += v
    else:
        continue # Skip items that would exceed the capacity

return solution, total_value, total_weight

```

Greedy by value

Another greedy strategy involves selecting items in descending order of value until the total exceeds the capacity.

```

import numpy as np

def greedy_knapsack_value(weights, values, capacity):
    """
    Greedy algorithm for the 0/1 Knapsack Problem based on value.

    Args:
        weights (np.ndarray): Weights of the items.
        values (np.ndarray): Values of the items.
        capacity (int): Capacity of the knapsack.

    Returns:
        tuple: Selected items (binary array), total value, total weight.
    """

    num_items = len(weights)

    # Create a list of items with their values and original indices
    items = list(zip(values, weights, range(num_items)))

    # Sort items by value in decreasing order
    items.sort(reverse=True)

```

```

total_weight = 0
total_value = 0
solution = np.zeros(num_items, dtype=int)

# Select items based on the sorted order
for v, w, idx in items:
    if total_weight + w <= capacity:
        solution[idx] = 1
        total_weight += w
        total_value += v
    else:
        continue # Skip items that would exceed the capacity

return solution, total_value, total_weight

```

Greedy by ratio

Based on value-to-weight ratio.

```

def greedy_knapsack_ratio(weights, values, capacity):
    """
    Greedy algorithm for the 0/1 Knapsack Problem based on value-to-weight ratio.

    Args:
        weights (np.ndarray): Weights of the items.
        values (np.ndarray): Values of the items.
        capacity (int): Capacity of the knapsack.

    Returns:
        tuple: Selected items (binary array), total value, total weight.
    """

    num_items = len(weights)

    # Calculate value-to-weight ratio for each item
    ratio = values / weights

    # Create a list of items with their ratios and original indices

```

```

items = list(zip(ratio, values, weights, range(num_items)))

# Sort items by ratio in decreasing order
items.sort(reverse=True)

total_weight = 0
total_value = 0
solution = np.zeros(num_items, dtype=int)

# Select items based on the sorted order
for r, v, w, idx in items:
    if total_weight + w <= capacity:
        solution[idx] = 1
        total_weight += w
        total_value += v
    else:
        continue # Skip items that would exceed the capacity

return solution, total_value, total_weight

```

Genetic Algorithm

See lecture notes for details.

```

import random

def initialize_population(pop_size, num_items):
    """
    Initialize the population with random binary strings.

    Args:
        pop_size (int): Number of individuals in the population.
        num_items (int): Number of items in the knapsack problem.

    Returns:
        np.ndarray: Initialized population.
    """
    return np.random.randint(2, size=(pop_size, num_items))

```

```

def evaluate_fitness(population, weights, values, capacity, penalty_factor=10):
    """
    Evaluate the fitness of each individual in the population.

    Args:
        population (np.ndarray): Current population.
        weights (np.ndarray): Weights of the items.
        values (np.ndarray): Values of the items.
        capacity (int): Capacity of the knapsack.
        penalty_factor (float): Penalty factor for exceeding capacity.

    Returns:
        np.ndarray: Fitness values for the population.
    """
    total_weights = np.dot(population, weights)
    total_values = np.dot(population, values)
    penalties = penalty_factor * np.maximum(0, total_weights - capacity)
    fitness = total_values - penalties
    return fitness

```

```

def tournament_selection(population, fitness, tournament_size):
    """
    Select individuals from the population using tournament selection.

    Args:
        population (np.ndarray): Current population.
        fitness (np.ndarray): Fitness values of the population.
        tournament_size (int): Number of individuals in each tournament.

    Returns:
        np.ndarray: Selected parents.
    """
    pop_size = population.shape[0]
    selected_indices = []
    for _ in range(pop_size):
        participants = np.random.choice(pop_size, tournament_size, replace=False)
        best = participants[np.argmax(fitness[participants])]
        selected_indices.append(best)
    return population[selected_indices]

```



```

def roulette_selection(population, fitness):
    """
    Select individuals from the population using roulette wheel selection.

    Args:
        population (np.ndarray): Current population.
        fitness (np.ndarray): Fitness values of the population.

    Returns:
        np.ndarray: Selected parents.
    """
    # Adjust fitness to be non-negative
    min_fitness = np.min(fitness)
    adjusted_fitness = fitness - min_fitness + 1e-6 # small epsilon to avoid zero division
    total_fitness = np.sum(adjusted_fitness)
    probabilities = adjusted_fitness / total_fitness
    pop_size = population.shape[0]
    selected_indices = np.random.choice(pop_size, size=pop_size, p=probabilities)
    return population[selected_indices]

```



```

def single_point_crossover(parents, crossover_rate):
    """
    Perform single-point crossover on the parents.

    Args:
        parents (np.ndarray): Selected parents.
        crossover_rate (float): Probability of crossover.

    Returns:
        np.ndarray: Offspring after crossover.
    """
    num_parents, num_genes = parents.shape
    np.random.shuffle(parents)
    offspring = []
    for i in range(0, num_parents, 2):
        parent1 = parents[i]
        parent2 = parents[i+1 if i+1 < num_parents else 0]
        child1 = parent1.copy()
        child2 = parent2.copy()
        if np.random.rand() < crossover_rate:

```

```

        point = np.random.randint(1, num_genes) # Crossover point
        child1[:point], child2[:point] = parent2[:point], parent1[:point]
        offspring.append(child1)
        offspring.append(child2)
    return np.array(offspring)

```

```

def uniform_crossover(parents, crossover_rate):
    """
    Perform uniform crossover on the parents.

    Args:
        parents (np.ndarray): Selected parents.
        crossover_rate (float): Probability of crossover.

    Returns:
        np.ndarray: Offspring after crossover.
    """
    num_parents, num_genes = parents.shape
    np.random.shuffle(parents)
    offspring = []
    for i in range(0, num_parents, 2):
        parent1 = parents[i]
        parent2 = parents[i+1 if i+1 < num_parents else 0]
        child1 = parent1.copy()
        child2 = parent2.copy()
        if np.random.rand() < crossover_rate:
            mask = np.random.randint(0, 2, size=num_genes).astype(bool)
            child1[mask], child2[mask] = parent2[mask], parent1[mask]
        offspring.append(child1)
        offspring.append(child2)
    return np.array(offspring)

```

```

def mutation(offspring, mutation_rate):
    """
    Apply bit-flip mutation to the offspring.

    Args:
        offspring (np.ndarray): Offspring after crossover.
        mutation_rate (float): Probability of mutation for each bit.
    """

```

Returns:

```
    np.ndarray: Offspring after mutation.
    """
    num_offspring, num_genes = offspring.shape
    mutation_matrix = np.random.rand(num_offspring, num_genes) < mutation_rate
    offspring[mutation_matrix] = 1 - offspring[mutation_matrix]
    return offspring
```

```
def elitism(population, fitness, elite_size):
```

"""

Preserve the top-performing individuals in the population.

Args:

```
    population (np.ndarray): Current population.
    fitness (np.ndarray): Fitness values of the population.
    elite_size (int): Number of top individuals to preserve.
```

Returns:

```
    np.ndarray: Elite individuals.
    """
    elite_indices = np.argsort(fitness)[-elite_size:] # Get indices of top individuals
    elites = population[elite_indices]
    return elites
```

```
def genetic_algorithm(weights, values, capacity, pop_size=100, num_generations=200, crossover_rate=0.5,
                      mutation_rate=0.05, elite_percent=0.02, selection_type='tournament', tournament_size=3,
                      crossover_type='single_point'):
```

"""

Main function to run the genetic algorithm for the 0/1 knapsack problem.

Args:

```
    weights (np.ndarray): Weights of the items.
    values (np.ndarray): Values of the items.
    capacity (int): Capacity of the knapsack.
    pop_size (int): Population size.
    num_generations (int): Number of generations.
    crossover_rate (float): Probability of crossover.
    mutation_rate (float): Probability of mutation.
    elite_percent (float): Percentage of elites to preserve.
    selection_type (str): 'tournament' or 'roulette'.
```

```

    tournament_size (int): Number of individuals in tournament selection.
    crossover_type (str): 'single_point' or 'uniform'.

Returns:
    tuple: Best solution, best value, and best weight found.
"""

num_items = len(weights)
elite_size = max(1, int(pop_size * elite_percent))
population = initialize_population(pop_size, num_items)

average_fitness_history = []
best_fitness_history = []

for generation in range(num_generations):
    fitness = evaluate_fitness(population, weights, values, capacity)

    # Track average and best fitness
    average_fitness = np.mean(fitness)
    best_fitness = np.max(fitness)
    average_fitness_history.append(average_fitness)
    best_fitness_history.append(best_fitness)

    # Elitism
    elites = elitism(population, fitness, elite_size)

    # Selection
    if selection_type == 'tournament':
        parents = tournament_selection(population, fitness, tournament_size)
    elif selection_type == 'roulette':
        parents = roulette_selection(population, fitness)
    else:
        raise ValueError("Invalid selection type")

    # Crossover
    if crossover_type == 'single_point':
        offspring = single_point_crossover(parents, crossover_rate)
    elif crossover_type == 'uniform':
        offspring = uniform_crossover(parents, crossover_rate)
    else:

```

```

        raise ValueError("Invalid crossover type")

    # Mutation
    offspring = mutation(offspring, mutation_rate)

    # Create new population
    population = np.vstack((elites, offspring))

    # Ensure population size
    if population.shape[0] > pop_size:
        population = population[:pop_size]
    elif population.shape[0] < pop_size:
        # Add random individuals to fill population
        num_new_individuals = pop_size - population.shape[0]
        new_individuals = initialize_population(num_new_individuals, num_items)
        population = np.vstack((population, new_individuals))

    # After all generations, return the best solution
    fitness = evaluate_fitness(population, weights, values, capacity)
    best_index = np.argmax(fitness)
    best_solution = population[best_index]
    best_value = np.dot(best_solution, values)
    best_weight = np.dot(best_solution, weights)

    return best_solution, best_value, best_weight, average_fitness_history, best_fitness_history

def genetic_algorithm_do_n(weights, values, capacity, pop_size=100, num_generations=200, crossover_type='tournament',
                           mutation_rate=0.05, elite_percent=0.02, selection_type='tournament', tournament_size=3, repeats=10):

    best_solution = None
    best_value = -1
    best_weight = -1

    best_averages = []
    best_bests = []

    for i in range(repeats):

        solution, value, weight, average_history, best_history = genetic_algorithm(weights, values, capacity, pop_size, num_generations, crossover_type,
                                           mutation_rate, elite_percent, selection_type, tournament_size)

```

```

        if value > best_value and weight <= capacity:
            best_solution = solution
            best_value = value
            best_weight = weight
            best_averages = average_history
            best_bests = best_history

    return best_solution, best_value, best_weight, best_averages, best_bests

```

Tests

Testing our genetic algorithm on data from [Google OR-Tools](#).

```

import matplotlib.pyplot as plt

def test_genetic_algorithm():
    # Sample data

    values = np.array([
        360, 83, 59, 130, 431, 67, 230, 52, 93, 125, 670, 892, 600, 38, 48, 147,
        78, 256, 63, 17, 120, 164, 432, 35, 92, 110, 22, 42, 50, 323, 514, 28,
        87, 73, 78, 15, 26, 78, 210, 36, 85, 189, 274, 43, 33, 10, 19, 389, 276,
        312])

    weights = np.array([
        7, 0, 30, 22, 80, 94, 11, 81, 70, 64, 59, 18, 0, 36, 3, 8, 15, 42, 9, 0,
        42, 47, 52, 32, 26, 48, 55, 6, 29, 84, 2, 4, 18, 56, 7, 29, 93, 44, 71,
        3, 86, 66, 31, 65, 0, 79, 20, 65, 52, 13])

    capacity = 850

    # Run genetic algorithm
    best_solution, best_value, best_weight, avg_fitness, best_fitness = genetic_algorithm_do_n(
        weights, values, capacity, pop_size=50, num_generations=100,
        crossover_rate=0.8, mutation_rate=0.05, elite_percent=0.02,
        selection_type='tournament', tournament_size=3)

    print("Best Solution:", best_solution)
    print("Best Value:", best_value)

```

```

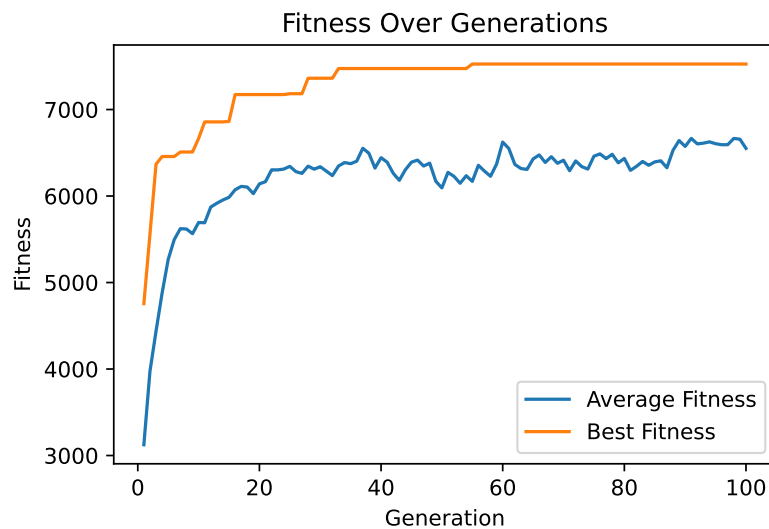
print("Best Weight:", best_weight)

# Plot the fitness over generations
generations = range(1, len(avg_fitness) + 1)
plt.plot(generations, avg_fitness, label='Average Fitness')
plt.plot(generations, best_fitness, label='Best Fitness')
plt.xlabel('Generation')
plt.ylabel('Fitness')
plt.title('Fitness Over Generations')
plt.legend()
plt.show()

test_genetic_algorithm()

```

Best Solution: [1 1 0 1 1 0 1 0 0 0 1 1 1 0 1 1 0 1 1 1 1 1 0 1 0 0 1 0 1 1 1 1 0 1 0 0
0 1 1 0 1 1 0 1 0 0 1 1 1]
Best Value: 7526
Best Weight: 848



Testing all the algorithms on data from pages.mtu.edu/~kreher/cages/data/knapsack/.

```

import pandas as pd

BASE_URL = 'https://pages.mtu.edu/~kreher/cages/data/knapsack/'

datasets = [
    'ks_8a.dat', 'ks_8b.dat', 'ks_8c.dat', 'ks_8d.dat', 'ks_8e.dat', 'ks_12a.dat',
    'ks_12b.dat', 'ks_12c.dat', 'ks_12d.dat', 'ks_12e.dat', 'ks_16a.dat', 'ks_16b.dat',
    'ks_16c.dat', 'ks_16d.dat', 'ks_16e.dat', 'ks_20a.dat', 'ks_20b.dat', 'ks_20c.dat',
    'ks_20d.dat', 'ks_20e.dat', 'ks_24a.dat', 'ks_24b.dat', 'ks_24c.dat', 'ks_24d.dat',
    'ks_24e.dat'
]

columns = [
    'file_path', 'capacity',
    'gw_value', 'gw_weight',
    'gv_value', 'gv_weight',
    'gr_value', 'gr_weight',
    'ga_value', 'ga_weight'
]

df = pd.DataFrame(columns=columns)

for idx, file_path in enumerate(datasets):

    values, weights, capacity = read_knapsack_data(BASE_URL + file_path)

    solution, total_value, total_weight = greedy_knapsack_weight(weights, values, capacity)

    gw_value = total_value
    gw_weight = total_weight

    solution, total_value, total_weight = greedy_knapsack_value(weights, values, capacity)

    gv_value = total_value
    gv_weight = total_weight

    solution, total_value, total_weight = greedy_knapsack_ratio(weights, values, capacity)

    gr_value = total_value
    gr_weight = total_weight

```



```

solution, total_value, total_weight, avg_fitness, best_fitness = genetic_algorithm_do_n(
    weights, values, capacity, pop_size=50, num_generations=100,
    crossover_rate=0.8, mutation_rate=0.05, elite_percent=0.02,
    selection_type='tournament', tournament_size=3, crossover_type='single_point')

ga_value = total_value
ga_weight = total_weight

df.loc[idx] = [
    file_path, capacity,
    gw_value, gw_weight,
    gv_value, gw_weight,
    gr_value, gr_weight,
    ga_value, ga_weight
]

df.to_csv("knapsack.csv", index=False)

print(df)

```

	file_path	capacity	gw_value	gw_weight	gv_value	gw_weight	gr_value	\
0	ks_8a.dat	1863633	874414	1803989	925369	1803989	925369	
1	ks_8b.dat	1822718	724029	1421763	836649	1421763	724029	
2	ks_8c.dat	1609419	771637	1609296	756847	1609296	713452	
3	ks_8d.dat	2112292	749458	1558340	1006793	1558340	881823	
4	ks_8e.dat	2493250	1224805	2386238	1300939	2386238	1300939	
5	ks_12a.dat	2805213	1180238	2323972	1409053	2323972	1381444	
6	ks_12b.dat	3259036	1334963	2639964	1681436	2639964	1602435	
7	ks_12c.dat	2489815	926226	1808471	1152681	1808471	1303224	
8	ks_12d.dat	3453702	1679959	3406646	1724265	3406646	1858992	
9	ks_12e.dat	2520392	1277814	2429214	1216398	2429214	1309915	
10	ks_16a.dat	3780355	1654432	3150713	1886539	3150713	2018230	
11	ks_16b.dat	4426945	1838356	3601726	2182562	3601726	2170190	
12	ks_16c.dat	4323280	1741661	3539978	2125245	3539978	2176322	
13	ks_16d.dat	4450938	2051218	4155271	2189910	4155271	2207441	
14	ks_16e.dat	3760429	1735397	3442535	1954173	3442535	1967510	
15	ks_20a.dat	5169647	2558243	5101533	2658865	5101533	2721946	
16	ks_20b.dat	4681373	2230065	4543967	2419141	4543967	2383424	
17	ks_20c.dat	5063791	2128763	4361690	2410432	4361690	2723135	

18	ks_20d.dat	4286641	1870486	3557405	2158431	3557405	2276327
19	ks_20e.dat	4476000	2115412	4173744	2159969	4173744	2294511
20	ks_24a.dat	6404180	2886589	5845661	3174264	5845661	3393387
21	ks_24b.dat	5971071	2961351	5941814	3019080	5941814	3164151
22	ks_24c.dat	5870470	2505304	5008038	2830470	5008038	3045772
23	ks_24d.dat	5762284	2711513	5247821	3047367	5247821	3135427
24	ks_24e.dat	6654569	3278044	6634696	3296337	6634696	3401688

	gr_weight	ga_value	ga_weight
0	1714834	925369	1714834
1	1421763	853704	1750340
2	1422422	771637	1609296
3	1682688	1084704	2059405
4	2377405	1300939	2377405
5	2672179	1468476	2804581
6	2953017	1753926	3254705
7	2406387	1329478	2458307
8	3412958	1858992	3412958
9	2477116	1309915	2477116
10	3768480	2018230	3768480
11	4071350	2311731	4392978
12	4054333	2282303	4315302
13	4245406	2298302	4422372
14	3616049	2030691	3755734
15	5054489	2788040	5161352
16	4471059	2471511	4676284
17	5029940	2723135	5029940
18	4273053	2280911	4275282
19	4353690	2350457	4471547
20	6379172	3393387	6379172
21	5911388	3194906	5970122
22	5820857	3066886	5848030
23	5734259	3150365	5754023
24	6435390	3501861	6649161

In the context of the 25 instances of the 0/1 knapsack problem, the genetic algorithm consistently outperformed the greedy algorithms. Specifically, it achieved solutions that were equivalent to the best greedy algorithm results in 8 cases and surpassed them in 17 cases, with up to 6% improvement.

Resources

- [DEAP](#) is an evolutionary computation framework designed for rapid prototyping and testing of ideas. This environment has been developed at Université Laval since (at least) 2012.
 - [Solving the knapsack problem using DEAP, complete example.](#)